Estimating Unit Costs of Nutrients from Market Prices of Feedstuffs

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ABSTRACT

The determinations of nutrient costs and breakeven prices of feedstuffs are important activities in the field of nutritional economics. Current methods are either extensions of the Petersen method, which dates back to 1932, or mathematical programming methods generally set as linear programming problems. Both sets of methods have severe limitations that make them unsuitable or biased for deriving unit costs of nutrients and aggregate breakeven costs of feedstuffs in a given market. We propose a least-squares method that, under verifiable conditions, yields maximum likelihood estimates of unit costs of nutrients and breakeven prices of feedstuffs. Assumptions to the maximum likelihood estimates are verified with studentized residuals, leverage values, and variance inflation factors. The method has been programmed into a Windows application and can be used by those with little formal training in statistics.
(**Key words:** nutrient costs, breakeven prices, Peterson method, maximum likelihood estimates)

Abbreviation key: BAR = ground barley, **CAM** = canola meal, **EBM** = equation-based methods, **GSC** = ground shelled corn, **IBM** = inequation-based methods, **LP** = linear programming, **ML** = maximum likelihood, **MT** = metric tonne, **PM** = Peterson method, **SBM** = dehulled, solvent-extracted soybean meal, **VIF** = variance inflation factors.

INTRODUCTION

A variety of methods have been proposed to estimate unit costs of nutrients and, implicitly or explicitly, the feeding value of feedstuffs. All methods fall into one of two general categories: equation-based (**EBM**) and inequation-based methods (**IBM**). For EBM, a set of equations developed from the nutritional composition of referee feeds is solved by using their market prices. The best-known method among this group is the Petersen Method (**PM**) in which the energy and protein compositions of corn grain and soybean meal are equated to their respective prices, setting a set of two equations with two unknowns. The method dates back to 1932 (14) and is presented and discussed at length by Morrison (11). Although widely used, the method is fundamentally flawed in that it assumes efficient markets in commodity trading and implies economically incoherent behavioral patterns by buyers and sellers of commodities.

The second series of methods, IBM, are basically constrained optimization models solved using mathematical programming techniques (1, 20). Linear programming (**LP**) is the best known member of this group and became widely used in animal nutrition with the discovery of an efficient algorithm (3) and the advent of high-speed computers. Within an LP model, a cost function is minimized subject to a series of inequations forcing the solution to meet the nutritional requirements of the animal for which the diet is being optimized. Linear programs suffer from being case specific, and they deliver little information on the unit costs of nutrients. Nutrients with nonbinding constraints have an implicit unit cost of zero. Shadow costs of binding nutrients provide information on unit costs that are valid only at the margin. Additionally, the information delivered has a very narrow inference range because it provides estimates that are sound only for one group of animals in a given herd. Consequently, LP is limited in providing estimates of aggregate unit costs of nutrients within a given market. This research attempts to circumvent these problems by developing a new procedure to provide estimates of aggregate unit costs of nutrients and breakeven prices of feedstuffs based on the trading of all feed commodities in a given market.

PROCEDURE DERIVATION

The Petersen Method

In the PM method (14), prices of n feedstuffs are equated to their composition in n nutrients. In most

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published applications, n has been set to 2 or 3. In the United States, TDN and CP have traditionally been used, although more recent applications have included other nutrients. In nearly all cases, US applications have used corn and soybean meal as the referee feeds. Without loss of generality, we can setup our equations using NE_L and CP for nutrients and ground, shelled corn (**GSC**) and dehulled, solvent extracted (48%) soybean meal (**SBM**) as the referee feeds. According to NRC (13), GSC and SBM contain 1.96 and 2.01 Mcal/kg NE_L, and 100.0 and 551.0 g/kg of CP on a DM basis, with a moisture content of 120 and 100 g/kg for GSC and SBM, respectively. Therefore, a metric tonne (**MT**) of GSC contains 1724.8 Mcal of NE_L and 88.0 kg of CP on an as-fed basis. Likewise, one MT of SBM contains 1809.0 Mcal of NE_L and 495.9 kg of CP. In the PM scheme, the following system of two equations with two unknowns is set:

$$\begin{aligned} GSC\$ &= 1724.8\ NE\$ + 88.0\ CP\$ \\ SBM\$ &= 1809.0\ NE\$ + 495.9\ CP\$ \end{aligned} \qquad [1]$$

Where

GSC$ = cost of GSC per MT,
SBM$ = cost of SBM per MT,
NE$ = cost per Mcal of NE_L (unknown), and
CP$ = cost per kg of CP (unknown).

This system of equations is easily solved. As an example, NE$ = 0.046 and CP$ = 0.236 when GSC$ and SBM$ are set at $100 and $200 per MT, respectively. These estimates of NE$ and CP$ can then be used to calculate the implicit values, often referred to as breakeven prices, of all other commodities in a given market.

Adding a nutrient to the estimation process is straightforward algebraically. An additional referee feed must be used for each additional nutrient. The software Feedval 4 from the University of Wisconsin, for example, sets six equations to solve for six nutrients. (Feedval 4 is available as a Microsoft Excel™ spreadsheet via the worldwide web at www.wisc.edu/dysci/vwex/nutritn/spreadshees/sprds.htm.) Referee feeds are blood meal, urea, corn grain, tallow, limestone, and dicalcium phosphate and are used to estimate the unit costs of Ca, P, RUP, RDP, fat, and TDN.

Although useful, the PM method contains fundamental flaws that cannot be ignored. First, referee feeds are never either well or poorly priced. That is, the method implicitly assumes that referee feeds are always priced at their breakeven prices. Invariably, one finds other commodities that are priced under their own breakeven prices. Hence, one would conclude that referee feeds should never be purchased based on a PM evaluation. This is an odd conclusion because the method implicitly assumes that referee feeds are market movers and set the prices of other commodities in the marketplace. The second flaw of the method is that PM assumes perfectly competitive markets. This implies that those trading referee feeds have perfect market information, with trading occurring at a perfect equilibrium point between supply and demand. Indirectly, this implies that the dairy industry must have a significant effect on the total demand of referee feeds (GSC and SBM in our example), a conclusion that is not supported by the current relative demand by the dairy industry for most feedstuffs. The dairy industry accounts for less than 10% of the total US demand for corn and SBM and for much less of the total demand for any of the reference feeds used by Feedval 4 (17). Third, the application of PM over a long period of time (years) results in an implicit incoherent economic behavior by buyers and sellers. That is, buyers keep purchasing some commodities well above their breakeven prices while sellers keep selling other commodities at prices considerably less than their breakeven prices. Fourth, it is difficult to augment PM to accommodate additional nutrients, not because of the algebra involved, but because of the difficulty in identifying proper referee feeds. With more than three or four nutrients involved, the system of equation tends to become ill-conditioned (4), with the consequence that a small change in the price of one referee feed results in substantial changes in the estimated costs of nutrients. This ill-conditioning characteristic of the equation system also applies to the nutritional composition coefficients. That is, one must assume near-perfect knowledge of the composition of referee feeds. This assumption may be reasonable for CP, a poor indicator of biological value in ruminants, but is greatly challenged when nutrients with more uncertain characteristics, such as RUP and NE_L are being considered.

Multiple Petersen Method

Some of the problems associated with the PM can be alleviated by considering more than two feedstuffs for the estimation of the unit cost of two nutrients. To understand how this can be accomplished, it helps to recast [1] using a more general notation as follows:

$$[2]\ Y_i = \sum_{j}^{2} X_{ij}B_j \quad i = 1,2$$

where

Y_i = price per MT of feedstuffs i,

X_{ij} = amount of nutrient j in a MT of feedstuffs i, and
B_j = cost per unit of nutrient j.

The equation set in [2] can be solved for any two feedstuffs (we will ignore here the obvious difficulty resulting from two feedstuffs of identical composition or, more generally, from collinearity between two equations i and i′ with $i \neq i'$). For example, we could evaluate the unit costs of NE_L and CP using GSC, SBM, ground barley (**BAR**), and canola meal (**CAM**), and use the standard nutritional composition reported by NRC (13), and price per MT of \$90, \$150, \$100, and \$200, respectively, for BAR, CAM, GSC and SBM. If we use BAR and CAM to set the first set of equations, we get:

$$\text{BAR:} \quad 90 = 1707.2\ \text{NE\$} + 118.8\ \text{CP\$} \qquad [3]$$

$$\text{CAM:} \quad 150 = 1582.4\ \text{NE\$} + 404.8\ \text{CP\$}$$

Likewise, a second set of equations can be set using BAR and GSC:

$$\text{BAR:} \quad 90 = 1707.2\ \text{NE\$} + 118.8\ \text{CP\$} \qquad [4]$$

$$\text{GSC:} \quad 100 = 1724.8\ \text{NE\$} + 88.0\ \text{CP\$}$$

With four feedstuffs, we can set six systems of paired equations and get six separate, although not independent, estimates of \$NE and \$CP. These estimates are reported in Table 1. Clearly, the selected pair of feedstuffs has a dramatic effect on the estimates. Across all pairs, mean estimates have SE that are approaching the magnitude of the means. Also, NE\$ and CP\$ estimates are highly correlated: the Pearson product-moment correlation (18) equals –0.85 ($P < 0.03$) and the Spearman rank correlation (18) equals –0.94 ($P < 0.005$).

Table 1. Estimated unit cost of NE_L and CP using the Multiple Petersen Method with ground barley (BAR), canola meal (CAM), ground shelled corn (GSC), and dehulled, solvent extracted soybean meal (SBM)[1].

Pair of Feedstuffs	NE_L (\$/Mcal)	CP (\$/kg)
BAR, CAM	0.037	0.226
BAR, GSC	0.724	–0.283
BAR, SBM	0.033	0.283
CAM, GSC	0.049	0.180
CAM, SBM	–0.125	0.861
GSC, SBM	0.046	0.236
Mean estimate	0.127	0.251
SEM	0.122	0.149
Median estimate	0.042	0.231

[1]Using the following cost per metric ton: BAR = \$90, CAM = \$150, GSC = \$100, and SBM = \$200.

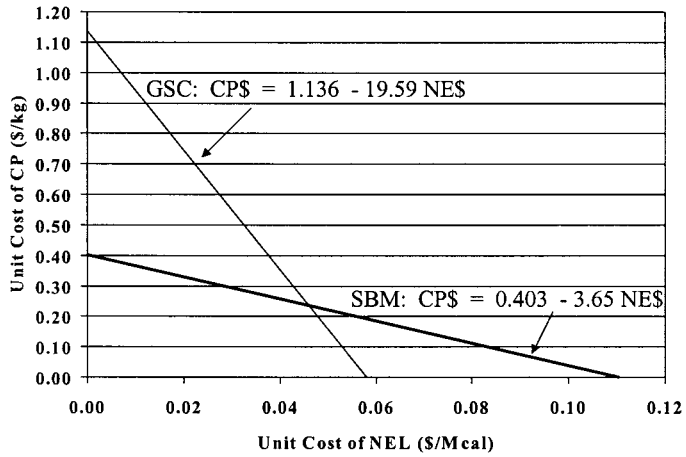

Figure 1. Graphical presentation of the relatively well-conditioned system of equations for ground, shelled corn (GSC) and dehulled, solvent extracted soybean meal (SBM) where unit cost of CP (CP\$) and unit cost of NE_L (NE\$) are estimated.

This method of estimation is fundamentally flawed because it assumes that all systems of paired equations have the same accuracy of estimation. This assumption ignores the fact that the condition number, defined as the ratio of the maximum to the minimum stretch induced by the X_{ij} coefficients, varies dramatically across systems of equations (4). For a system of two equations, the condition number can be easily presented in graphical form. Figure 1 shows the system of equations for GSC and SBM. Although not orthogonal, the two lines have slopes that are sufficiently different (GSC = –19.58, SBM = –3.65) for a relatively accurate estimate of their point of intersection. This result is in contrast with the system of equations of CAM and SBM presented in Figure 2. The graph of the two equations shows that they are nearly parallel lines (slope of CAM = –3.91), so that moving one of the two lines a little would drastically alter their point of intersection. This last system of equations has a high condition number and the system is ill-conditioned, meaning that the solution is very sensitive to small changes in the data (composition or price).

Although the multiple Petersen method appears ill-suited for the task of estimating unit costs of nutrients, we chose to present it because it bridges the concepts behind PM and the maximum likelihood method (**ML**) that we have developed.

Maximum Likelihood Method

Two important changes must be made in our conceptual approach to the problem. First, we must move away from the concept of a few referee feedstuffs. We shall instead use an approach in which all feedstuffs are

used jointly and simultaneously to estimate unit costs of nutrients. From a market point of view, this implies that no feedstuffs rule the market. Second, we will no longer assume efficient markets. That is, we will implicitly recognize that markets are not instantaneously self-correcting. At any point in time, any feedstuff can be priced over or under its breakeven price. This is accomplished by adding an error term to equation [2] as follows:

$$Y_i = \sum_{j}^{2} X_{ij}B_j + e_i \qquad [5]$$

This equation can be easily expanded to more than two nutrients:

$$\mathbf{Y} = \mathbf{XB} + E \qquad [6]$$

where

Y is an n × 1 vector of prices for n feedstuffs,
X is an n × m matrix of coefficients of m nutrients for n feedstuffs,
B is an m × 1 vector of unit costs of m nutrients, and
E is an n × 1 vector of errors.

The system of equations [6] has an infinite number of solutions because it contains more unknowns than the number of equations (n unknown e_i and m unknown B_j in a system of n equations). The system, however, has a unique solution if we set for objectives to minimize the sum of squares of deviations (i.e., minimize $\sum_i e_i^2$). In matrix notation, this implies the minimization of:

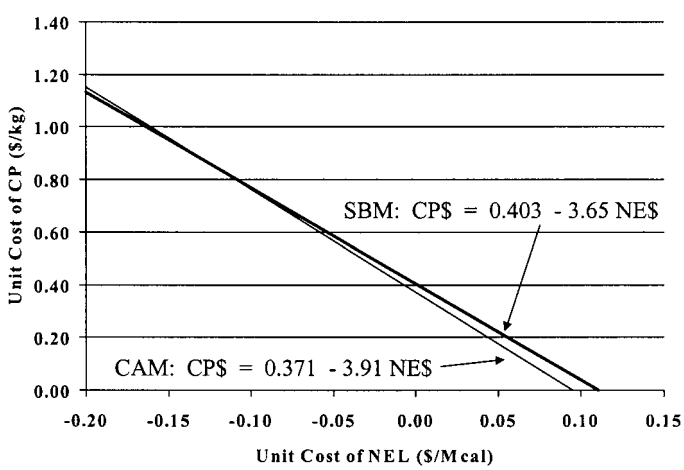

Figure 2. Graphical representation of the ill-conditioned system of equations for canola meal (CAM) and dehulled, solvent extracted soybean meal (SBM) where unit cost of CP (CP$) and unit cost of NE_L (NE$) are estimated.

$$\mathbf{E'E} = \mathbf{Y'Y} - 2\mathbf{B'X'Y} + \mathbf{B'X'XB} \qquad [7]$$

Differentiating [7] with respect to **B** and setting the resulting matrix equation equal to a vector of zeros, the least-squares estimate of **B** is the vector **b**:

$$\mathbf{b} = (\mathbf{X'X})^{-1}\mathbf{X'Y} \qquad [8]$$

This solution **b** has a unique set of properties. First, **b** is an estimate of **B** that minimizes the error sum of squares **E′E** regardless of any distribution properties of the errors. Second, the elements of **b** are linear functions of Y_1, Y_2, …, Y_n and provide unbiased estimates of **B**. These estimates have the minimum variances among any estimates that are linear functions of the Y's, irrespective of distribution properties of the errors. Last, and importantly, if the errors are independent and identically distributed from a normal distribution with a mean of 0 and a variance of σ^2, i.e., $E \approx N(\mathbf{O}, \mathbf{I}\sigma^2)$, then **b** is the ML estimate of **B**. Therefore, equation [8] produces ML estimates of the unit costs of nutrients using market prices and composition of feedstuffs if the residuals are independent, have common variance and are all distributed according to a normal distribution with a mean of zero and a variance of σ^2. Consequently, it will be important in the application of our ML method to ensure that these assumptions are realistic.

A direct benefit of using a ML approach is that various statistics can be calculated using conventional, well-known statistical methods (5). In particular, an estimate of the variance (σ^2) is:

$$\begin{aligned}\sigma^2 &= \mathbf{Y'Y} - \mathbf{Y'X}\,(\mathbf{X'X})^{-1}\mathbf{X'Y}/n - m\\ &= \mathbf{Y'Y} - \mathbf{Y'Xb}/n - m\\ &= \mathbf{Y'Y} - \mathbf{b'X'Y}/n - m\end{aligned} \qquad [9]$$

The variance of estimated unit costs of nutrients **b** is:

$$\mathrm{Var}(\mathbf{b}) = (\mathbf{X'X})^{-1}\sigma^2 \qquad [10]$$

Lastly, the variance of the predicted breakeven price of a feedstuffs is given by:

$$\mathrm{Var}(\hat{Y}_i) = \mathbf{X_i}(\mathbf{X'X})^{-1}\mathbf{X_i}\sigma^2 \qquad [11]$$

Figure 3 reports the results obtained when using the ML method on our simple example consisting of BAR, CAM, GSC, and SBM. Although purposely kept simple, this example illustrates some of the appealing and unique features of a ML approach: estimates of unit costs of nutrients have standard errors, correlations between estimates is explicit, and range as opposed to point estimates of breakeven prices of feedstuffs can be calculated relatively easily.

$$X = \begin{bmatrix} 1707.2 & 118.8 \\ 1582.4 & 404.8 \\ 1724.8 & 88.0 \\ 1809.0 & 495.9 \end{bmatrix} \qquad Y = \begin{bmatrix} 90 \\ 150 \\ 100 \\ 200 \end{bmatrix}$$

$$X'X = \begin{bmatrix} 1.16660E7 & 1.8922E6 \\ 1.18922E6 & 4.3164E5 \end{bmatrix} \qquad X'Y = \begin{bmatrix} 9.2529E5 \\ 1.7939E5 \end{bmatrix}$$

$$(X'X)^{-1} = \begin{bmatrix} 2.9668E-7 & -1.3006E-6 \\ -1.3006E-6 & 8.0184E-6 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.0412 \\ 0.2350 \end{bmatrix} \qquad VAR(b) = \begin{bmatrix} 4.7868E-5 & 2.0985E-4 \\ 2.0985E-4 & 1.2937E-3 \end{bmatrix}$$

$$r_{b_1, b_2} = 0.843$$

$$\hat{Y} = \begin{bmatrix} 98.25 \\ 160.32 \\ 91.74 \\ 191.07 \end{bmatrix} \qquad VAR(\hat{Y}) = \begin{bmatrix} 72.65 \\ 63.02 \\ 88.70 \\ 98.26 \end{bmatrix}$$

Figure 3. Matrices involved in the least-squares, maximum likelihood estimation of the unit cost of NE_L (NE$) and the unit cost of CP (CP$), using ground barley, canola meal, ground shelled corn, and dehulled, solvent extracted soybean meal. Composition used was from NRC (13). (Prices were set at $90, $150, $100, and $200 per metric ton for ground barley, canola meal, ground shelled corn, and soybean meal, respectively.)

ESTIMATION PROBLEMS

The Intercept Question

The inclusion of an intercept term in model [6] allows the estimation of an overall average residual, i.e., a number equal to the average value of feedstuffs after the value of the nutrients have been accounted for. In general, this term should be close to zero. Failure to include an intercept term when indeed one is present in the population of feedstuffs being studied leads to biased estimates of the other elements of **b**. In our application (19), we recommend to first use a model that includes an intercept, to test for the significance of such intercept, and to remove the intercept term when it is nonsignificant before reestimating **b**.

Outliers

The term outlier refers to an observation that is extreme relative to the bulk of the remaining observations. The presence of one or more outliers invalidates the assumption of normal distribution of errors in [6], resulting in an estimate of **b** that is no longer ML. Many different rules have been proposed for identifying and rejecting outliers (2, 15, 16). However, the automatic rejection of outliers is not always a very wise procedure. At times, the outlier is giving information not provided by other observations. In our application (19), we chose not to include an automatic rejection scheme of possible outliers. The user is provided with studentized residual measures, and observations with studentized residuals

that exceed ± 2.5 standard deviations should be looked at closely to ensure that the nutritional content of the feedstuffs is realistic and that the market price used is real.

Leveraged Observations

The detection of leveraged observations is a topic that receives little coverage, although leveraged observations can result in substantial problems in parameter estimation (10). A leverage point is an outlier among the predictor variables. Leverage points can be identified statistically from the leverage values h_i, which are the diagonal elements of the projection matrix (8):

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad [12]$$

Thus, we can rewrite [11] as follows:

$$\mathrm{Var}(\hat{Y}_i) = h_i\sigma^2 \quad [13]$$

Expressed this way, the effect of the leverage value on the variance of the estimate of the breakeven price for feedstuffs i is clearly seen.

In our application (19), feedstuffs with leverage values $h_i > 2\ (p + 1)/n$, where p = number of parameters and n = number of feedstuffs, are deemed leveraged and their inclusion in the data set should be questioned. Feedstuffs with leveraged values $h_i > 3\ (p + 1)/n$ should almost invariably be removed from the dataset.

Collinearity

Collinearities are approximate linear dependencies among the columns of predictor variable values. Collinearities can be due to model constraints, population characteristics that restrict variate values, sampling deficiencies, or overdefined models. Collinearities produce large coefficient estimates and inflate the variance of parameter estimates (9). In short, collinearity induces ill-conditioning of the $\mathbf{X}'\mathbf{X}$ matrix, that is, the determinant of $\mathbf{X}'\mathbf{X}$ is close to zero. This condition can be thought of in two ways: either, the model is overspecified (we are trying to estimate the cost of more nutrients than are needed to express the breakeven prices of the feedstuffs selected) or the data are not adequate to estimate all of the selected nutrients.

Collinearities can be measured with variance inflation factors (**VIF**) which are the diagonal elements of $(\mathbf{X}'\mathbf{X})^{-1}$ when $\mathbf{X}'\mathbf{X}$ is scaled to correlation form. In general, estimability is acceptable when all VIF < 10 (by definition, VIF = 1 in the absence of collinearity). In instances where VIF > 20, action must be taken. Additional feedstuffs with different nutritional characteristics can be included in the data, and this may result in

Table 2. Composition of feedstuffs and Midwest market prices in June 1999 used for the estimation of the unit cost of NDF, RUP, RDP, and NE_L[1].

	NDF (%)	RUP (%)	RDP (%)	NE_L (Mcal/kg)	Cost ($/metric tonne)	Location
Barley, grain	16.720	3.929	7.951	1.705	87.50	Kansas City
Blood meal	0.828	66.387	14.573	1.377	345.00	Minneapolis
Brewers grains, dehydrated	42.320	15.395	7.973	1.377	72.00	Chicago
Corn gluten meal, 60% CP	12.600	35.683	24.797	1.854	210.00	Chicago
Corn gluten feed	40.500	5.760	17.280	1.722	55.00	Chicago
Corn grain, ground	7.920	4.869	3.931	1.727	78.57	Chicago
Soybean meal, solvent-extracted, 44% CP	13.261	15.544	28.867	1.725	125.00	Chicago
Beet pulp, dehydrated	49.140	5.064	3.763	1.630	125.00	Chicago
Molasses, sugarcane	0.000	0.000	4.350	1.233	65.00	Minneapolis
Canola meal, solvent-extracted	24.752	11.878	25.068	1.429	96.00	Minneapolis
Cottonseed, whole	40.480	6.441	14.719	2.053	191.00	Chicago
Cottonseed meal, 41% CP	23.660	17.843	23.653	1.585	142.00	Chicago
Distillers dried grains	40.480	12.650	10.350	1.873	85.00	Chicago
Feather meal	36.387	56.036	24.015	1.488	170.00	Minneapolis
Fish meal, Menhaden	1.840	36.818	24.546	1.535	450.00	Chicago
Hominy feed	49.500	5.435	4.915	1.810	72.00	Chicago
Meat and bone meal	3.720	24.653	25.660	1.506	170.00	Chicago
Poultry byproduct meal	5.640	34.122	27.918	1.661	200.00	Kansas City
Soybean meal, solvent-extracted, 48% CP	7.200	17.357	32.234	1.810	136.00	Chicago
Tallow	0.000	0.000	0.000	5.840	200.00	Chicago
Wheat middlings	32.930	3.742	12.634	1.616	48.00	Chicago
Soybean hulls	60.970	4.625	6.386	1.608	60.00	Chicago
Bakery waste, dehydrated	16.560	2.412	7.432	1.896	70.00	Minneapolis
Cottonseed hulls	81.900	1.119	2.612	0.894	67.50	Chicago
Wheat grain, ground	10.680	3.275	10.965	1.812	95.00	Kansas City

[1]Composition data taken from NRC (12, 13) and Midwest market prices taken from Feedstuffs (6) are reported on an as-is basis.

Midwest – June 1999 – Initial Estimates

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Main results				
R-square	Adj. R-square	Root MSE	Dep Mean	C.V.
0.876	0.853	63.406	136.622	46.410

Analysis of variance					
Source	DF	Sum of square	Mean square	F Value	Prob>F
Model	4	597660.881	149415.220	37.165	0.0000
Error	21	84426.642	4020.316		
Total	25	682087.524			

Inference of coefficients					
Nutrient name	Estimate	Standard error	t-statistic	Prob.> |T|	Var. inflation
NDF	-0.027937	0.025907	-1.078	0.2931	1.7861
RUP	0.198323	0.044633	4.443	0.0002	2.6895
RDP	0.010147	0.073512	0.138	0.8915	4.1449
NEl	0.053797	0.013701	3.926	0.0008	3.1499

Correlation Matrix			
1	2	3	4
1.000	0.069	-0.043	-0.498
	1.000	-0.655	-0.007
		1.000	-0.494
			1.000

Calibration set							
Name	Actual [/T]	Predicted [/T]	Predicted-Actu	Leverage	Student res.	75.0% CI	75.0% CI
Barley, grain	87.498	91.086	3.588	0.046	-0.058	74.936	107.236
Blood meal	345.000	333.031	-11.969	* 0.552	0.282	277.306	388.756
Brewers grains, dehy.	72.000	106.249	34.249	0.083	-0.564	84.663	127.836
Corn gluten meal, 60%	210.000	230.031	20.031	0.103	-0.334	205.970	254.093
Corn gluten feed	55.000	87.771	32.771	0.095	-0.543	64.643	110.900
Corn grain, ground	78.570	99.969	21.399	0.071	-0.350	79.970	119.968
Soybean meal, solv, 44%	125.000	144.280	19.280	0.166	-0.333	113.716	174.844
Beet pulp, dehydrated	125.000	72.934	-52.066	0.113	0.872	47.703	98.164
Molasses, sugarcane	65.000	61.057	-3.943	0.045	0.064	45.230	76.885
Canola meal, solv-extd	96.000	108.119	12.119	0.134	-0.205	80.611	135.627
Cottonseed, Whole	191.000	106.132	-84.868	0.081	1.396	84.759	127.505
Cottonseed meal, 41%	142.000	139.719	-2.281	0.086	0.038	117.772	161.666
Distillers dried grains	85.000	121.073	36.073	0.070	-0.590	101.245	140.900
Feather meal	170.000	279.433	109.433	0.298	-2.060	238.466	320.400
Fish meal - Menhaden	450.000	224.906	-225.094	0.121	3.787	198.788	251.024
Hominy feed	72.000	83.248	11.248	0.114	-0.188	57.928	108.568
Meat and bone meal	170.000	174.420	4.420	0.108	-0.074	149.774	199.066
Poultry byproduct meal	200.000	218.896	18.896	0.126	-0.319	192.313	245.479
Soybean meal, solv, 48%	136.000	159.714	23.714	0.223	-0.424	124.320	195.108
Tallow	200.000	203.722	3.722	** 0.670	-0.102	142.338	265.106
Wheat Middlings	48.000	77.863	29.863	0.059	-0.486	59.565	96.161
Soybean Hulls	60.000	64.025	4.025	0.156	-0.069	34.358	93.692
Bakery waste, dehy	69.998	94.336	24.338	0.065	-0.397	75.257	113.416
Cotton, hulls	67.500	2.843	-64.657	0.352	1.267	-41.657	47.342
Wheat grain, ground	94.991	97.683	2.691	0.064	-0.044	78.774	116.591

* Feedstuffs with a high leverage on estimates.

** Feedstuffs with a very high leverage on estimates

SESAME: Regression results

Figure 4. Initial table of results as reported by SESAME (19) for the estimation of unit costs of NDF, RDP, RUP, and NE_L using market prices for 25 Midwest feedstuffs (6). This figure is an integral copy of the output from SESAME. Abbreviations are as follows: Adj R-square = adjusted R^2, Root MSE = square root of the mean squared error, Dep Mean = mean value of the dependent variable, C.V. = coefficient of variation, DF = degrees of freedom, Prob = probability, Var. inflation = variance inflation factor, Actual [/T] = observed price per metric ton, Predicted [/T] = predicted price per metric ton, Predicted-Actu = predicted minus observed price per metric ton, Leverage = leverage values, Student res. = studentized residuals, and 75.0% CI = lower and higher 75% confidence limits on predicted price.

Midwest – June 1999 – Final Estimates

18.12.1999 14:19:06

Main results				
R-square	Adj. R-square	Root MSE	Dep Mean	C.V.
0.965	0.956	23.722	102.978	23.036

Analysis of variance					
Source	DF	Sum of square	Mean square	F Value	Prob>F
Model	4	246177.477	61544.369	109.363	0.0000
Error	16	9004.046	562.753		
Total	20	255181.524			

Inference of coefficients					
Nutrient name	Estimate	Standard error	t-statistic	Prob.> |T|	Var. inflation
NDF	-0.002430	0.012544	-0.194	0.8488	2.6597
RUP	0.194114	0.040025	4.850	0.0002	5.2668
RDP	0.007716	0.043497	0.177	0.8614	8.2045
NEl	0.038849	0.009392	4.136	0.0008	6.9475

Correlation Matrix			
1	2	3	4
1.000	0.026	0.268	-0.719
	1.000	-0.713	-0.019
		1.000	-0.565
			1.000

Calibration set							
Name	Actual [/T]	Predicted [/T]	Predicted-Actu	Leverage	Student res.	75.0% CI	75.0% CI
Barley, grain	87.498	75.771	-11.727	0.122	0.528	65.874	85.669
Brewers grains, dehy.	72.000	107.478	35.478	0.174	-1.646	95.657	119.299
Corn gluten meal, 60%	210.000	207.103	-2.897	0.407	0.159	189.038	225.167
Corn gluten feed	55.000	83.753	28.753	0.151	-1.316	72.746	94.761
Corn grain, ground	78.570	79.989	1.419	0.233	-0.068	66.310	93.668
Soybean meal, solv, 44%	125.000	124.943	-0.057	0.212	0.003	111.895	137.992
Beet pulp, dehydrated	125.000	75.291	-49.709	0.138	2.257	64.777	85.805
Molasses, sugarcane	65.000	44.126	-20.874	0.142	0.950	33.464	54.787
Canola meal, solv-extd	96.000	99.147	3.147	0.212	-0.149	86.098	112.196
Cottonseed meal, 41%	142.000	127.640	-14.360	0.105	0.640	118.469	136.811
Distillers dried grains	85.000	114.754	29.754	0.114	-1.333	105.193	124.316
Hominy feed	72.000	83.254	11.254	0.145	-0.513	72.473	94.034
Meat and bone meal	170.000	152.571	-17.429	0.154	0.799	141.469	163.673
Poultry byproduct meal	200.000	195.026	-4.974	0.321	0.254	178.991	211.062
Soybean meal, solv, 48%	136.000	135.809	-0.191	0.278	0.009	120.883	150.735
Wheat Middlings	48.000	71.821	23.821	0.097	-1.057	63.010	80.632
Soybean Hulls	60.000	72.629	12.629	0.171	-0.585	60.927	84.332
Bakery waste, dehy	69.998	76.514	6.515	0.187	-0.305	64.281	88.747
Cotton, hulls	67.500	32.277	-35.223	0.470	2.039	12.868	51.686
Wheat grain, ground	94.991	77.750	-17.242	0.168	0.797	66.151	89.348

* Feedstuffs with a high leverage on estimates.

** Feedstuffs with a very high leverage on estimates

SESAME: Regression results

Figure 5. Final table of results as reported by SESAME (19) or the estimation of unit costs of NDF, RDP, RUP and NE_L using market prices for 20 Midwest feedstuffs (6). This figure is an integral copy of the output from SESAME. Abbreviations are as follows: Adj R-square = adjusted R^2, Root MSE = square root of the mean squared error, Dep Mean = mean value of the dependent variable, C.V. = coefficient of variation, DF = degrees of freedom, Prob = probability, Var. inflation = variance inflation factor, Actual [/T] = observed price per metric ton, Predicted [/T] = predicted price per metric ton, Predicted-Actu = predicted minus observed price per metric ton, Leverage = leverage values, Student res. = studentized residuals, and 75.0% CI = lower and higher 75% confidence limits on predicted price.

reduced collinearities. Alternatively, the nutrient associated with the largest VIF can be removed from the model. The remaining parameters must then be reestimated using the reduced **X** matrix.

EXAMPLE

Table 2 reports the composition of 25 feedstuffs (12, 13) and Midwest market prices (6) in June 1999. The nutrients selected for the estimation were: NE_L, RDP, RUP, and NDF. Results of the first estimation, as reported by a specialized Windows application, SESAME (19), are shown in Figure 4. The intercept estimate was not significant (B_0 = 40.67, SE = 71.54, P = 0.58) and was removed from the model in a preliminary step. All VIF were less than 5.0, indicating a well-conditioned **X′X** matrix. Of the 25 feedstuffs selected, tallow showed an excessive leverage value and was removed from the calibration set. Likewise, fishmeal had a large studentized residual and was also removed from the calibration set. In a subsequent iteration, the studentized residuals or leverage values of blood meal, feather meal, and whole cottonseed became excessive, resulting in their deletion from the calibration set.

Results from the final iteration are shown in Figure 5. The model explains a large proportion of the cost variance (uncorrected for the mean) with an $R^2 = 0.965$. Estimates of nutrient costs were as follows: NDF, –$0.002 (±0.013) per kg; RUP, $0.194 (±0.040) per kg; RDP, $0.008 (±0.043) per kg; and NE_L, $0.039 (±0.009) per Mcal. Both NDF and RDP costs were not different from zero and could be removed from the model. The strong negative correlations between RUP and RDP costs and between NDF and NE_L costs explain the magnitude of VIF, although all VIF were below a value of 10. SESAME calculates various statistics for all feedstuffs in the calibration set. These include: breakeven prices (labeled as "Predicted"), residual values (labeled as "Predicted-Actual"), leverage values (labeled as "Leverage"), studentized residuals (labeled "Student res."),

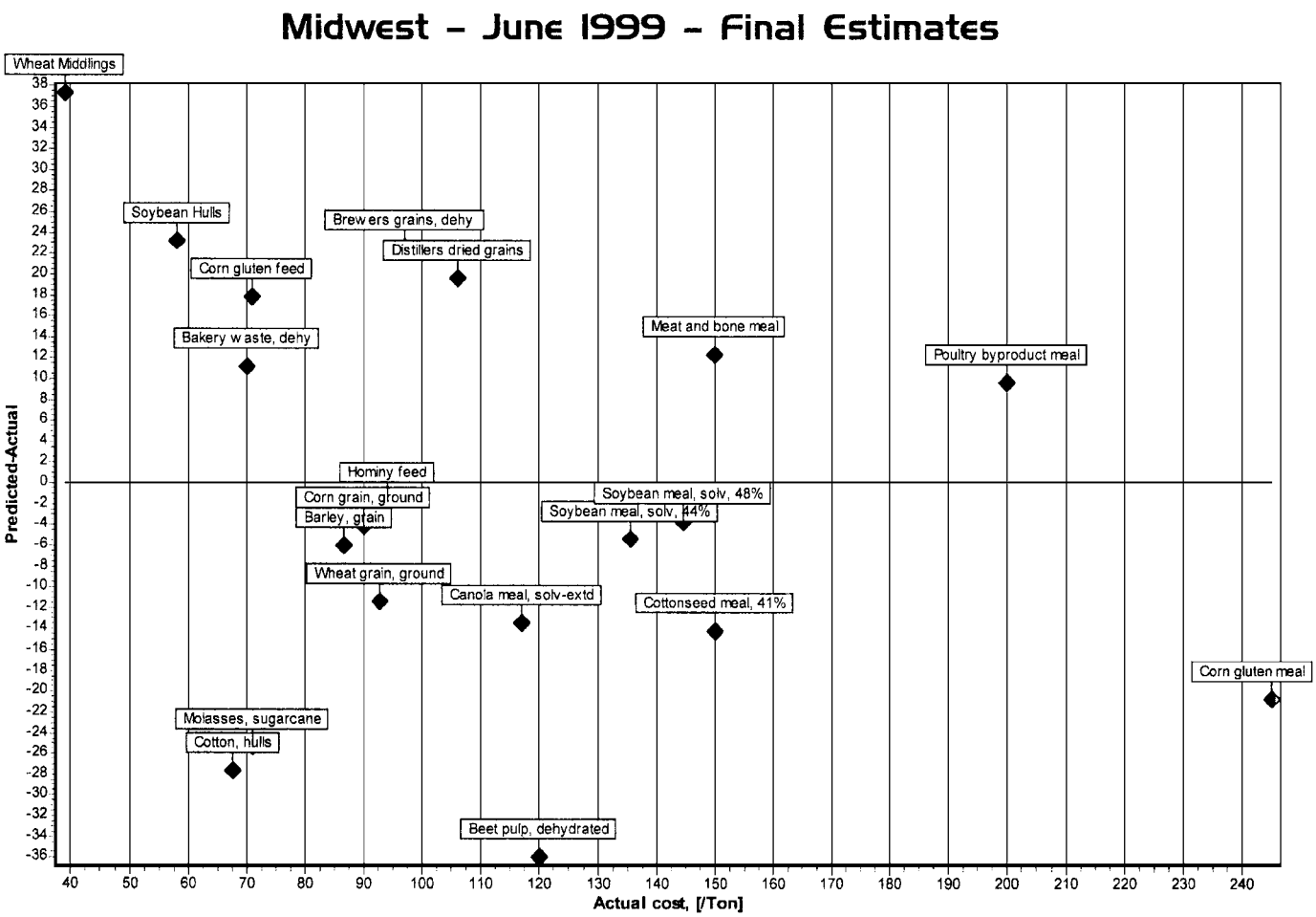

Figure 6. Residual plot from the estimation of unit costs of NDF, RDP, RUP, and NE_L for 20 Midwest feedstuffs (6).

and lower and upper 75% confidence interval of the breakeven prices (labeled "75% CI"). Deviations from estimated breakeven prices are also presented in a simple X-Y graphic of residuals versus actual costs of feedstuffs (Figure 6). Feedstuffs appearing below the zero residual line are relatively overpriced, whereas those appearing above the line are relatively underpriced.

CONCLUSIONS

The ML method uses the price of all feedstuffs traded in a given market to estimate the implicit costs of nutrients. Because it is a statistically based method, ML provides measures of dispersion of estimated nutrient costs and breakeven prices. Also, because the ML method does not use referee feeds, each feedstuffs used in the estimation can potentially have a breakeven price above or below its market price. The method has been programmed into a Windows application available from the authors. The long-run properties of ML estimates compared to PM remain to be determined.

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